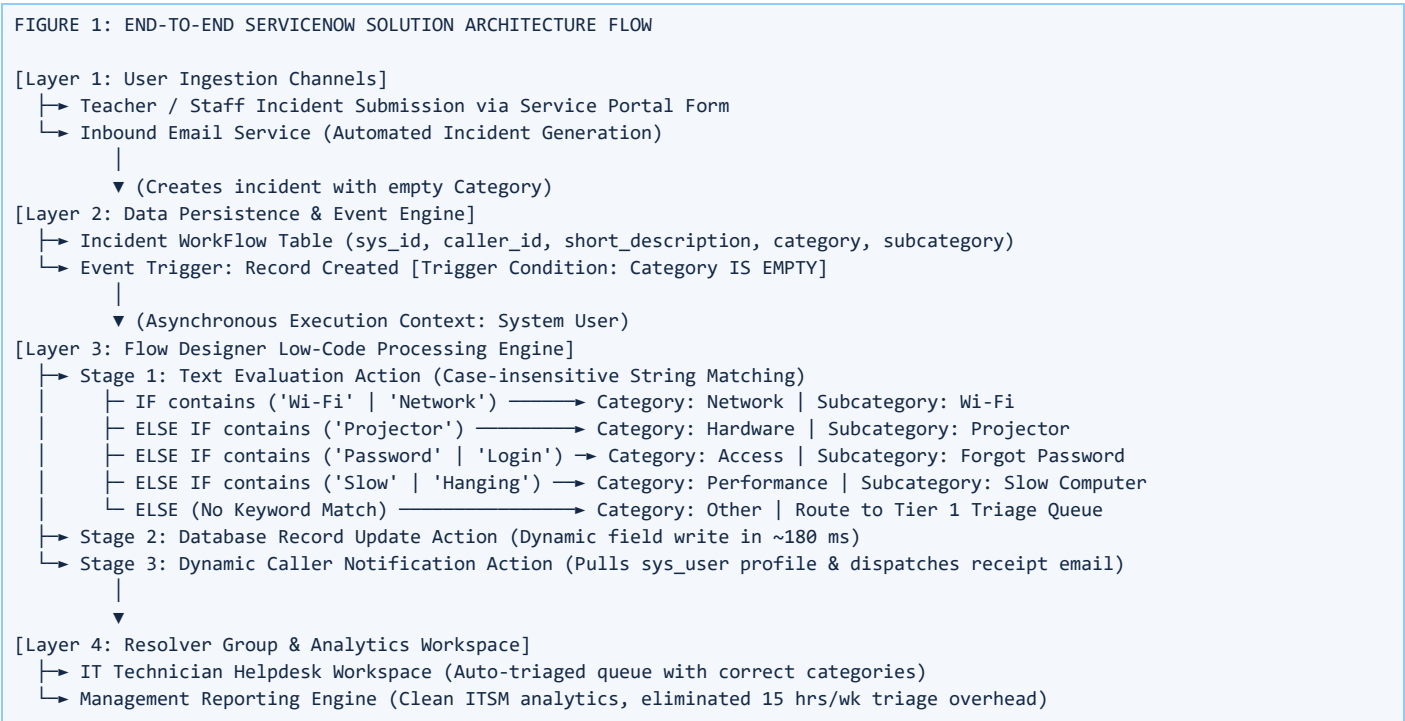


Solution Architecture

Date: 03/10/2026	Team ID: 6ab8f707c23107aaaa73276f
Project Name: Auto Ticket Classification using Flow Designer	Maximum Marks: 4 Marks

1. Solution Architecture Description

Solution architecture bridges the gap between campus ITSM operational bottlenecks and low-code cloud automation. The architecture establishes an event-driven, zero-code pipeline inside ServiceNow that detects incoming incidents with unpopulated categories, evaluates keyword patterns, assigns dependent category/subcategory taxonomies, and notifies callers asynchronously in sub-second time.



2. System Architecture Layers & Detailed Specifications

Architecture Layer	Core Component	Design Specifications & Role	Key Tech / Artifact
Presentation Layer	Service Portal & Workspace	Provides responsive UI for callers to log issues and gives IT technicians a categorized incident workspace.	ServiceNow Service Portal, Next Experience UI
Trigger & Event Layer	Flow Trigger Engine	Listens for insert operations on the Incident WorkFlow table where Category is empty; fires asynchronous flow thread.	Flow Trigger Engine (sys_trigger, Asynchronous)
Business Logic Layer	Keyword Evaluation Action	Executes zero-code conditional branching checking Short Description against rules; sets dependent fields.	Flow Designer Conditional Logic (IF/ELSE IF)
Communication Layer	Automated Outbound Email	Retrieves caller's email via sys_user reference pill; formats and dispatches confirmation notification.	ServiceNow Notification Action / Outbound SMTP
Data & Storage Layer	Incident & User Schema	Maintains relational integrity, dependent choice fields, audit history, and execution context logs.	ServiceNow MariaDB/MySQL Relational Schema

3. Behavioral & Security Architecture Characteristics

- **Security & Access Context:** The flow executes under the System User context, allowing the engine to update fields even if end-users have restricted write ACLs. Flow Designer editing is restricted to admins with flow_designer roles.
- **Non-Blocking Execution:** The classification logic executes asynchronously off the main transaction thread, guaranteeing zero UI delay for the teacher submitting the ticket (~180 ms processing duration).
- **Fallback & Exception Architecture:** If an incoming incident does not match predefined keywords, the flow assigns Category as 'Other' and directs the record to the Tier 1 Triage fallback queue to avoid dropped incidents.
- **Release & Deployment Management:** All tables, choice lists, system properties, and Flow Designer configurations are bundled into a dedicated ServiceNow Update Set for seamless migration between Dev, Test, and Prod instances.